Reg. No. | | | | | | | | | | | [18MTASD221]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

INTRODUCTION TO AERO ELASTICITY

Time: 03 Hours **Maximum Marks:100**

Instructions: (i) Answer five full questions.
(ii) Question numbers one and two are compulsory.
(iii) Missing data may be suitably assumed.

1. a. A vibrating system has amass of 1 kg suspended by a spring of stiffness 900 N/m . It also has a viscous damper with a damping constant of 30Ns/m. The system is subjected to a harmonic excitation of force $F = F_0 \cos\omega t$ with $F_0 = 10N$. Calculate the following:

i.	Undamped natural frequency ω_n	**01**
ii.	Damped natural frequency ω_d	**01**
iii.	Amplitude and phase angle at $\omega=0$	**02**
iv.	Phase angle at $\omega=0.5\omega_n$, $\omega=\omega_n$, $\omega=1.5\omega_n$ and $\omega=3\omega_n$	**04**
v.	Amplitude Ratio at $\omega=0.5\omega_n$, $\omega=\omega_n$, $\omega=1.5\omega_n$ and $\omega=3\omega_n$	**04**
vi.	Maximum amplitude and corresponding ω and phase angle	**03**
vii.	Show a plot of response amplitude and phase angle variation as ω/ω_n varies from 0 to ∞.	**05**

2. a. Define; **2+2**

i. Control Effectiveness and

ii. Control Reversal.

b. Derive an expression for the aileron reversal speed for a simple 2-D wing with aileron. The 2-D wing of chord length c is in an uniform air flow of speed U and density ρ at an initial angle of attack of α_0. The 2-D wing is restrained in torsion by a torsional spring of Spring Constant K_θ at its elastic axis location at a distance 'e' from its aerodynamic centre. Consider the 2-D wing to be in equilibrium at its initial angle of attack of $\alpha_0$andconsideronly changes in aerodynamic lift L and moment about aerodynamic centre M_{AC} produced due to an aileron deflection δ.(Draw a neat sketch of the wing section showing all the parameters). (ii) Show plots of (L/L_R) versus (q/q_R) for q_D/q_R= 0.5, 1.0 and 1.5, where q is dynamic pressure and q_D are q_R its values at reversal and divergence. **10+6**

3. a. Consider a simple 2-DOF system of an airfoil of chord length c=2b in a uniform flow of velocity U and air density ρ. The airfoil is restrained in lateral motion 'h' by a linear spring K_h and in torsional motion 'α' by a torsional spring K_α at **10+10**

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distance (ba) behind the mid-chord position. The mass of the airfoil is 'm' and its centre of gravity (CG) is located at a distance (bx_α) behind the point of restraint. I_α is the mass moment of inertia about the point of restraint and I_0 is the mass moment of inertia about the CG. Derive (i) the equations of equilibrium and (ii) conditions for flutter ,assuming for low speed flow, the unsteady lift L and moment M_y acting at the point of restraint are given by

$L= -\pi\rho b^3\omega^2\{L_h(h/b)+ [L_\alpha-L_h(1/2+a)]\alpha\}$

$M_y= -\pi\rho b^4\omega^2\{[M_h- L_h(1/2+a)](h/b)+ [M_\alpha-(L_\alpha+ M_h)(1/2+a)+ L_h(1/2+a)^2]\alpha\}$

OR

4 a. Define (i)Eigen values and (ii)Eigen Vectors with reference to a 2-DOF system. **4+2**

4 b. Find the Eigen values and Eigen Vectors of a coupled pendulum shown in Fig. 1 for m=1kg, K=100N/m, a= 0.6m and b=0.8m **14**

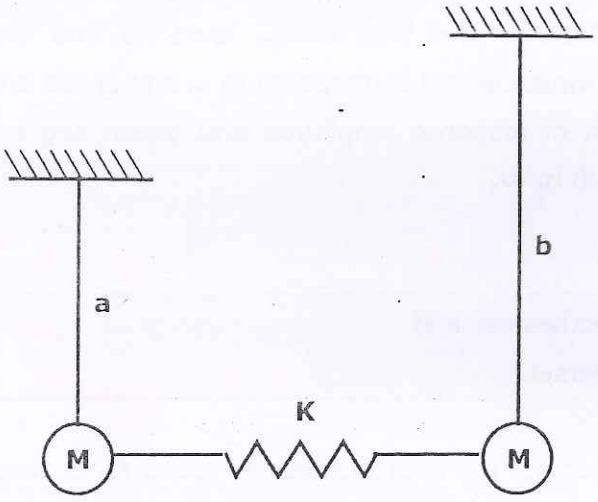

Fig. 1 Coupled Pendulum

M = 1Kg, K = 100N/m, a=0.6m, b=0.8m

5 a. i. Define Self Excited Oscillation. **3+3**

ii. Give 3 examples.

5 b. Define the possible instability scenarios for a SDOF spring-mass- viscous damper system when **7+7**

i. External Force F(t) is proportional to displacement x(t)

ii. External Force F(t) is proportional to velocity dx(t)/dt.

OR

6 a. Define torsional divergence of a wing. **04**

6 b. i. Derive an expression for the ratio of the total equilibrium angle of attack of an elastic wing to that of a rigid wing, $(\alpha_0+\theta)/\alpha_0$ for a typical section **14+2**

airfoil as a function of (q/q_D) where α_0 is the initial angle of attack and θ is the angle of twist about elastic axis, q is the dynamic pressure and q_D is the divergence dynamic pressure. The airfoil has a chord length c and is in an uniform air flow of speed U and density ρ. The airfoil section is restrained in torsion by a torsional spring of Spring Constant K_θ at its elastic axis location at a distance 'e' from its aerodynamic centre (draw a neat sketch). The wing section aerodynamic lift is L and aerodynamic moment about the aerodynamic centre is M_{AC}.

ii. Plot the variation of the ratio with (q/q_D).

7. **i.** Derive an expression for the lift efficiency (L/L_R) of a large AR rectangular straight wing of semi-span 'l', with constant chord length c and constant torsional stiffness GJ in a uniform airflow of speed U and density ρ, for the case when the attitude of the rigid airplane is specified(i.e., $\alpha_0$atwing root is specified). The distance from the line of elastic axis of the wing to its line of aerodynamic centres is 'e' . Assume a symmetric airfoil section and the line of centre of gravity coinciding with line of elastic axis. Assume strip theory aerodynamics. **16+4**

ii. Show variation of lift efficiency with q/q_D where q is the dynamic pressure and q_D, its value at divergence rn.

OR

8. Write Short Notes on any four: **4X5**

i. Collar's Aeroelastic Triangle

ii. Dynamic load problems in aeroelasticity

iii. Quality factor and Band width

iv. Normal modes and Mode normalization method

v. Buffeting

vi. Viscous damping